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Subject: Data Engineering

Practical 3: MongoDB Operations

Aim:

To perform single and multiple document insertion and retrieval using MongoDB shell (mongosh).

Step 1: Insert Single Document

Query:

```
db.items.insertOne({ name: "laptop", price: 999 });
```

Output:

```
>_MONGOSH
> db.items.insertOne({ name: "laptop", price: 999 });
< {
  acknowledged: true,
  insertedId: ObjectId('6ac649038fa08d8f4561c384')
}
test>
```

Step 2: View Inserted Document

Query:

```
db.items.find();
```

Output:

```
> db.items.find();
< {
  _id: ObjectId('6ac51e1cd4bbf249c034346a'),
  name: 'laptop',
  price: 999
}
{
  _id: ObjectId('6ac649038fa08d8f4561c384'),
  name: 'laptop',
  price: 999
}
test> |
```

Step 3: Insert Multiple Documents

Query:

```
db.products.insertMany([
  { name: "phone", price: 500, stock: 10 },
  { name: "tablet", price: 300, stock: 5 },
  { name: "watch", price: 150, stock: 0 }
]);
```

Output:

```
> db.products.insertMany([
  { name: "phone", price: 500, stock: 10 },
  { name: "tablet", price: 300, stock: 5 },
  { name: "watch", price: 150, stock: 0 }
]);
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6ac6498b8fa08d8f4561c385'),
    '1': ObjectId('6ac6498b8fa08d8f4561c386'),
    '2': ObjectId('6ac6498b8fa08d8f4561c387')
  }
}
test>
```

Step 4: View Products in Formatted JSON

Query:

```
db.products.find().pretty();
```

Output:

```
> db.products.find().pretty();
< {
  _id: ObjectId('6ac51e43d4bbf249c034346b'),
  name: 'phone',
  price: 500,
  stock: 10
}
{
  _id: ObjectId('6ac51e43d4bbf249c034346c'),
  name: 'tablet',
  price: 300,
  stock: 5
}
{
  _id: ObjectId('6ac51e43d4bbf249c034346d'),
  name: 'watch',
  price: 150,
  stock: 0
}
{
  _id: ObjectId('6ac6498b8fa08d8f4561c385'),
  name: 'phone',
  price: 500,
  stock: 10
}
```

```
{
  _id: ObjectId('6ac6498b8fa08d8f4561c386'),
  name: 'tablet',
  price: 300,
  stock: 5
}
{
  _id: ObjectId('6ac6498b8fa08d8f4561c387'),
  name: 'watch',
  price: 150,
  stock: 0
}
test>
```